

Poisson Distribution

Reading Objectives:

In this reading, you will understand the Poisson probability distribution with its corresponding probabilities, expectation, and its properties.

Main Reading Section:

Definition 1: A random variable X , taking on one of the values $0, 1, 2, \dots$, is said to be a Poisson random variable with parameter $\lambda, \lambda > 0$, if its probability mass function is given by

$$P\{X = i\} = e^{-\lambda} \frac{\lambda^i}{i!}, \quad i = 0, 1, \dots$$

The symbol e stands for a constant approximately equal to 2.7183.

Properties:

Let us first determine its moment-generating function:

$$\begin{aligned}\Phi(t) &= E[e^{-tX}] \\ &= \sum_{i=0}^{\infty} e^{ti} e^{-\lambda} \frac{\lambda^i}{i!} \\ &= e^{-\lambda} \sum_{i=0}^{\infty} e^{ti} \frac{\lambda^i}{i!} \\ &= e^{-\lambda} e^{\lambda e^t} \\ &= \exp\{\lambda(e^t - 1)\}\end{aligned}$$

Differentiation yields

$$\begin{aligned}\Phi'(t) &= \lambda e^t \exp\{\lambda(e^t - 1)\} \\ \Phi''(t) &= (\lambda e^t)^2 \exp\{\lambda(e^t - 1)\} + \lambda e^t \exp\{\lambda(e^t - 1)\}\end{aligned}$$

Evaluating at $t = 0$ gives that

$$\text{Mean: } E[X] = \Phi'(0) = \lambda$$

$$\text{Variance: } \text{Var}(X) = \Phi''(0) - (E[X])^2 = \lambda^2 + \lambda - \lambda^2 = \lambda$$

Recurrence Relation:

If X is Poisson with mean λ , then

$$\frac{P\{X=i+1\}}{P\{X=i\}} = \frac{e^{-\lambda} \lambda^{i+1}/(i+1)!}{e^{-\lambda} \lambda^i/i!} = \frac{\lambda}{i+1}$$

Relation between Binomial and Poisson distributions

The Poisson random variable has a wide range of applications in a variety of areas because it may be used as an approximation for a binomial random variable with parameters (n, p) when n is large and p is small. To see this, suppose that X is a binomial random variable with parameters (n, p) and let $\lambda = np$. Then

$$\begin{aligned}P\{X = i\} &= \frac{n!}{(n-i)! i!} p^i (1-p)^{n-i} \\ &= \frac{n!}{(n-i)! i!} \left(\frac{\lambda}{n}\right)^i \left(1 - \frac{\lambda}{n}\right)^{n-i} \\ &= \frac{n(n-1)\dots(n-i+1)}{n^i} \frac{\lambda^i}{i!} \frac{(1-\frac{\lambda}{n})^n}{(1-\frac{\lambda}{n})^i}\end{aligned}$$

Now, for n large and p small,

$$\left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda} \frac{n(n-1)\dots(n-i+1)}{n^i} \approx 1 \cdot \left(1 - \frac{\lambda}{n}\right)^i \approx 1$$

Hence, for n large and p small,

$$P\{X = i\} \approx e^{-\lambda} \frac{\lambda^i}{i!}$$

Example 1. Suppose that the average number of accidents occurring weekly on a particular stretch of a highway equals 3. Calculate the probability that there is at least one accident this week.

Solution: Let X denote the number of accidents occurring on the stretch of highway in question during this week. Because it is reasonable to suppose that there are a large number of cars passing along that stretch, each having a small probability of being involved in an accident, the number of such accidents should be approximately Poisson distributed. Hence,

$$\begin{aligned}P\{X \geq 1\} &= 1 - P\{X = 0\} \\ &= 1 - e^{-3} \frac{3^0}{0!} \\ &\approx 0.9502\end{aligned}$$